MSCI 342 Advanced Spreadsheet Modelling 2024

Emissions Modelling

With around 1.5 billion cars in the world today, vehicle emissions are a highly relevant topic, in terms of climate change, pollution, electric vehicles usage, safety, health and noise. A supplied datafile by the Vehicle Certification Agency (VCA) provides fuel efficiency and emissions data for over 50 manufacturers, for vehicles tested between 2015 to 2023.

Part A – General Analysis [20 marks]

1. How many total cars have been tested by the VCA in the supplied dataset?
2. How many petrol[1] cars have been tested?
3. How many diesel[2] cars have been tested?
4. How many electric[3] cars have been tested?
5. How many hybrid[4] cars have been tested?
6. Which manufacturer[5] has the most cars tested?
7. Which manufacturer[5] has the most petrol[1] cars tested?
8. Which manufacturer[5] has the most hybrid[4] cars tested?
9. How many manufacturers have had *only* petrol[1] cars tested?
10. How many manufacturers have had *only* hybrid[4] cars tested?

Part B – Audi Analysis [20 marks]

11. How many Audi cars have been tested?
12. How many diesel[2] Audi cars have been tested?
13. What is the average MPG[6] value for Audi cars, rounded to 2 decimal places?
14. What is the average MPG[6] value for diesel Audi cars, rounded to 2 decimal places?
15. What is the median CO_2 value for Audi cars, rounded to the nearest integer?
16. How many Audi cars are missing a numerical MPG value?
17. How many Audi cars are missing a numerical CO_2 value?
18. What is the highest CO_2 for an individual Audi car?
19. What is the model for the individual Audi car with the highest CO_2?
20. What is the fuel type for the individual Audi car with the lowest CO_2?

Part C – Analysis by Year [10 marks]

21. Which year saw the most cars tested?
22. Which year saw the biggest increase in the number of tests compared to the previous year?
23. Which year saw the biggest increase in tests for hybrid[4] cars compared to the previous year?
24. Which year has the highest number of non-numerical MPG values?
25. The highest average CO_2 for any manufacturer occurs in which year?

Part D – Extremes and Gears [10 marks]

26. How many different types of transmission (code) are there?
27. Which type of transmission (code) is the most numerous in the dataset?
28. How many cars feature a transmission code which does <u>not</u> contain a number?
29. Which type of transmission (code) produces the maximum power, on average, for petrol[1] cars?
30. What is the maximum power for an automatic[7] transmission car?

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1 petrol-only fuel, not hybrids or combinations
2 diesel-only fuel, not hybrids or combinations
3 electricity-only cars, not hybrids or combinations of fuels
4 a hybrid is defined as a *combination* of fuel types, for example 'Diesel Electric', or 'Petrol Hybrid'
5 if there is a *tie* for the number tested, then present the first manufacturer alphabetically
6 for all MPG calculations, use the 'Imperial Combined' figure
7 an automatic car has an "A" in the transmission code (upper or lower case) EXCEPT the following types: M6-AWD, MPS6AWD, MPS6-AWD and N/A

Part E – Fleet Costing [20 marks]

The sheet 'Fleet Car Database' contains a database of 44 journeys made by a fleet of cars. Based on the fuel economy values[6] in the main VCA dataset for each fleet car (manufacturer, model, year and fuel type) estimate the amount of fuel used for each journey, and the subsequent fuel cost.

Note 1: If there are multiple versions of a particular model listed in the VCA dataset, then use the average MPG value[6] across all the versions of that particular model, year and fuel type.
Note 2: Include a facility to add *new* journeys to the database, based on a user selection of the following items: a car in the fleet, a driver (code) and a mileage.
Note 3: Include a graphical display of the fuel and mileage data for each member of staff
Note 4: Ensure all your calculations will **update** when new journeys are added to the database, up to a maximum of 250 journeys.
Note 5: Include the facility to add a car to the fleet, up to a maximum of 5 additional cars.

31. What is the total fuel cost for the 44 journeys, to the nearest pence, based on the fuel prices given on the 'Fleet Cars' sheet?
32. Is there a particular fleet car that is preferred by female drivers? If so, which car (code)?

Model Design [20 marks]

High marks will go to models which follow the design principles of the module, as laid out in the documentation and described during the lectures. The model should be intuitive to use, well-structured, and not be more complicated than it needs to be, particularly in terms of data management.
The model should also, if practical, be *dynamic* in terms of its calculations – in other words if any changes are made to the data, then the model should be capable of updating the results/calculations in response.

Data Collection

The data itself is held on moodle in a folder called 'Coursework Data'.
Each student will receive a unique dataset, based on a *code number*, also listed on moodle
You **must** use the correct dataset, as all are different.
Failure to do so will generate incorrect answers, and incur a **mark penalty**.

Submission

Deliverable is an **Excel 2021 workbook** addressing the tasks, uploaded to moodle.
The model rules are as follows. Failure to comply with these rules will result in a mark penalty

- ✓ The model should open on the User sheet
- ✓ All answers should be presented in the correct cells on the User sheet
- ✓ The User sheet should also include your ID number
- ✓ Only correct results will receive marks
- ✓ All answers must be function-based
  i.e. **not** simple/static values, **not** generated by sorting/filtering, and **not** found manually by the user
- ✓ The model will be assessed on a university-spec. PC
- ✗ The model should not contain external links
- ✗ The model should not have any hidden sheets
- ✗ The model should not have any circular references
- ✗ The model should not exceed **20MB** in size

Deadline is: 23rd April 2024